



Institute of Science and Technology (IST)

Department of CSE

1st year 2nd Semester

Model Test 2022

Course Title: Statistics and Probability(CSE-510227)

Total Marks: 80, Time:3 Hours

[Answer any four questions]

- 1a) Discuss the concepts of data? Write the different data used in statistics. 3
- b) Write the different steps of a questionnaire. 2
- c) Why needed graphical representation of data? What are various types of charts and diagrams used in presenting statistical data? Explain one of them. 5
- c) The price of (thustk) of 35 computers are given bellow: 10

20 30 45 55 36 56 62 18 20 65 35 69 22 39 49 50 53

67 65 58 37 18 65 48 16 45 19 53 23 58 30 49 45 56 45

- i) Construct a frequency distribution from the above data.
- ii) Construct a stem and leaf diagram for this data
- iii) Draw a frequency polygon and also show the histogram.
- 2a) Define weighed arithmetic mean? Discuss the importance of weighed arithmetic mean. 3
- b) For two observations x & y prove that $A.M. \geq G.M. \geq H.M.$ 3

c) The following table gives the daily wage distribution of the employees of a commercial organization:

Profits per shop	250-300	300-350	350-400	400-450	450-500	500-550
No. of shops	20	10	11	25	14	30

- i) Compute arithmetic mean and 6th deciles. 10
- ii) Draw a histogram and then find modal value? 4
- 3a) When use skewness? Write its types and uses. 4
- b) Prove that for two observations prove that standard deviation is half of the range. 3
- c) The mean and standard deviation of 15 observations are found to be 10 and 5 respectively. On rechecking it was found that one of the observation with value 8 was incorrect. Calculate the correct mean and standard deviation if the correct observation value was 23? 8
- d) The total marks scored by two students Sathya and Vidhya in 5 subjects are 460 and 480 with standard deviation 4.6 and 2.4 respectively. Who is more consistent in performance? 5
- 4a) What is coefficient of variation? Write the difference between standard deviation and coefficient of variation. 4
- b) The following table given: 10

Daily income (tk)	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90
No. of workers	4	6	10	12	7	3	2	8	3

Calculate the standard deviation and coefficient of variation.

c) Display the data in a box plot, 12,5,6,12,6,14,16

5 a) Explain partial and biserial correlation ?

b) What is rank correlation? Formulate rank correlation formula.

c) Find coefficient of correlation and correlation ratio price and sales from the following data:

Price (in taka)	103	98	85	92	90	84	88	90	93	95
Sales (units)	500	610	700	630	670	800	800	750	700	680

6a) What is 3-variables regression ? How you estimate regression parameters ?

b) What is OLS method ? Write properties of estimators?

c) The following information :

	Advertisement expenditure(Tk lakhs)(x)	Sales (Tk crores)(y)
Average	45	98
standard deviation	12	23

Coefficient of correlation $r=0.91$.

i) Find the regression equations x on y and y on x .

ii) Find the likely sales when advertisement expenditure is 63 lakhs and predict the sales.

7a) What is association of attributes? Write different method of association of attribute. When we use contingency table?

b) How you measure of association for nominal data?

c) In an examination, at which 600 candidates appeared, boys outnumbered girls by 16% of all candidates. Number of passed exceeded the number of failed candidates by 310. Boys failing in the examination numbered 88. Find the Yule's coefficient of association between male and success in examination.